Mark Schachner

120 Malott Hall, Cornell University, Ithaca, NY | mjs623@cornell.edu | mark-schachner.com

Education

Cornell University — August 2021 – May 2027 (expected)
Ph.D., Mathematics — *Ithaca, NY*

Cornell University — August 2021 – May 2024
Master of Science, Computer Science — *Ithaca, NY*

- Relevant coursework: Analysis of Algorithms, Theory of Computation, Foundations of Programming Languages, Category Theory for Computer Scientists

The University of Chicago — September 2017 – June 2021
Bachelor of Science with Honors, Mathematics — *Chicago, IL*

Publications and Preprints

The group generated by a graph's derangements — October 2026
Preprint, arXiv:2610.18463 — *with Gabe Udell*

Computational techniques for sheaf cohomology of locally profinite sets — February 2026
Preprint, arXiv:2602.02976

Awards and Fellowships

Graduate Student Teaching Award — December 2025
$750

Graduate Research Assistantships — May – August 2023, May – August 2024
$18,547

NSF Graduate Research Fellowship — August 2021 – Present
$138,000

Experience

Teaching Assistant / Counselor — July 2020 – August 2021
Bridge to Enter Advanced Mathematics — *Los Angeles, CA*

- Summer outreach program to help underserved students enter advanced study in mathematics
- Responsibilities included supervising problem sessions and program activities, as well as working directly with students to develop critical thinking and creative problem-solving skills

Teaching

Teaching Assistant, MATH 1300 — August 2026 – Present
Cornell Mathematics Department — *Ithaca, NY*

- In-class teaching assistant for MATH 1300 Mathematical Explorations

Adjunct Faculty, Cornell Prison Education Program — January 2026 – May 2026
Cayuga Community College — *Auburn, NY*

- Instructor of record for MATH 112 Contemporary Mathematics
- Developed and taught weekly lessons in mathematics to incarcerated students at Cayuga Correctional Facility

Instructor, Ithaca High School Math Seminar — September 2025 – June 2026
Ithaca High School — *Ithaca, NY*

- Instructor for mathematics seminar
- Developed curriculum for high school students grades 10-12
- Proposed and supervised independent student projects in combinatory logic and infinitary voting theory

Instructor, Cornell Prison Education Program — January 2025 – February 2025
Cornell Mathematics Department — *Ithaca, NY*

- Instructor of record for MATH 112 Contemporary Mathematics (co-taught with Aria Beaupré)
- Developed and taught weekly lessons in mathematics to incarcerated students
- Session cut short after one month

Instructor of Record, MATH 1120 — January 2024 – May 2024
Cornell Mathematics Department — *Ithaca, NY*

- Instructor of record for MATH 1120 Calculus II

Recitation Teaching Assistant, MATH 2940 — September 2023 – December 2023
Cornell Mathematics Department — *Ithaca, NY*

- Teaching assistant for MATH 2940 Linear Algebra for Engineers
- Gave weekly recitations to students using active learning techniques

Recitation Teaching Assistant, MATH 2210 January 2023 – May 2023
Cornell Mathematics Department *Ithaca, NY*
- Teaching assistant for MATH 2210 Linear Algebra
- Gave weekly recitations to students using active learning techniques

Recitation Teaching Assistant, MATH 1910 September 2022 – December 2022
Cornell Mathematics Department *Ithaca, NY*
- Teaching assistant for MATH 1910 Calculus for Engineers
- Gave weekly recitations to students using active learning techniques

Service and Outreach

Vice President, Association for Women in Mathematics September 2026 – Present
Cornell University Mathematics Department *Ithaca, NY*
- Previously served as Outreach Chair and on Professional Development teams
- Planned, organized, and led Julia Robinson Math Festival, a day-long math outreach workshop
- Organized and led Mathematical Puzzles workshop for Cornell's Expanding Your Horizons event

Facilitator, TA Training August 2024 – August 2026
Cornell University Mathematics Department *Ithaca, NY*
- Collaborated with lecturers and fellow graduate students to facilitate multiple TA training sessions for incoming graduate students
- Gave sample discussion sections as a teaching model

Mentor, Directed Reading Program September 2022 – Present
Cornell University Mathematics Department *Ithaca, NY*
- Mentored several students in topics from mathematical foundations and set theory

Mentor, Nexus Scholars Program May 2023 – August 2023
Cornell University *Ithaca, NY*
- Mentored a research team advised by Justin Moore

Organizer, Ph.D. Mentoring Program May 2022 – May 2023
Cornell University Mathematics Department *Ithaca, NY*
- Collaborated with Sara Stephens to coordinate department-wide program to match matriculating Ph.D. students with experienced student mentors
- Additionally served as mentor in program

Instructor, Little Circle January 2023 – May 2023
Cornell University Mathematics Department *Ithaca, NY*
- With Alexander Vidinas, developed and presented weekly lessons in departmental outreach program

Talks

Toward an Accessible Mathematics Classroom IV. Back to Basics
Cornell Teaching Seminar 30 September 2026

Toward an Accessible Mathematics Classroom III. Tools and Rules
Cornell Teaching Seminar 1 October 2025

Toward an Accessible Mathematics Classroom II. Learning Disability
Cornell Teaching Seminar 26 March 2025

Toward an Accessible Mathematics Classroom
Cornell Teaching Seminar 29 November 2023

Test-giving & Test-taking
Cornell Teaching Seminar 22 March 2023

Professional Development

Early Graduate Teaching Cohort
Center for Teaching Innovation, Cornell University 10–17 April 2026

Researcher, Computer Assisted Proofs in Applied Mathematics
Simons Laufer Mathematical Sciences Institute (SLMath) 7–18 July 2025

Technical Skills

Languages: C, JavaScript, Haskell, Python, LaTeX, HTML, CSS, MATLAB